

Supplementary Tables. Hyperparameter Optimization and Empirical Selection

SUPPLEMENTARY TABLE 1. EFFECTS OF RNAP SEQUENCE LENGTH AND LEARNING RATE ON MODEL PERFORMANCE

Configuration	ACC	Precision	Recall	F1-score	AUC
81bp,lr=5e-5	0.811	0.829	0.794	0.804	0.897
81bp,lr=1e-4	0.822	0.848	0.814	0.820	0.916
81bp,lr=2e-4	0.826	0.868	0.805	0.810	0.923
162bp,lr=5e-5	0.842	0.855	0.835	0.839	0.923
162bp,lr=1e-4	0.851	0.877	0.829	0.843	0.939
162bp,lr=2e-4	0.854	0.881	0.834	0.844	0.947
243bp,lr=5e-5	0.855	0.857	0.861	0.855	0.931
243bp,lr=1e-4	0.862	0.881	0.848	0.856	0.945
243bp,lr=2e-4	0.861	0.879	0.854	0.855	0.952

SUPPLEMENTARY TABLE 2. COMPARISON OF DIFFERENT K-MER SCHEMES FOR PROMOTER SEQUENCE ENCODING

Configuration	k-mer	ACC	Precision	Recall	F1-score	AUC
162bp,lr=1e-4	2	0.845	0.862	0.833	0.839	0.930
	3	0.851	0.877	0.829	0.843	0.939
	4	0.862	0.886	0.842	0.854	0.950
	5	0.863	0.888	0.841	0.853	0.952
	6	0.863	0.881	0.848	0.858	0.947
	2	0.862	0.865	0.871	0.861	0.941
243bp,lr=1e-4	3	0.862	0.881	0.848	0.856	0.945
	4	0.868	0.882	0.862	0.863	0.954
	5	0.873	0.885	0.867	0.869	0.955
	6	0.867	0.882	0.858	0.862	0.952

SUPPLEMENTARY TABLE 3. PERFORMANCE COMPARISON OF TRANSFORMER PARAMETER COMBINATIONS

Configuration	ACC	Precision	Recall	F1-score	AUC
depth=3,heads=5,dim_head=16, attn_dropout=0.1,ff_dropout=0.1	0.878	0.900	0.859	0.872	0.959
depth=5,heads=5,dim_head=16, attn_dropout=0.1,ff_dropout=0.1	0.877	0.900	0.856	0.871	0.961
depth=4,heads=5,dim_head=16, attn_dropout=0.1,ff_dropout=0.1	0.879	0.893	0.871	0.876	0.960
depth=4,heads=3,dim_head=16, attn_dropout=0.1,ff_dropout=0.1	0.878	0.899	0.863	0.873	0.959
depth=4,heads=4,dim_head=16, attn_dropout=0.1,ff_dropout=0.1	0.877	0.891	0.869	0.874	0.960
depth=4,heads=6,dim_head=16, attn_dropout=0.1,ff_dropout=0.1	0.880	0.893	0.875	0.876	0.960
depth=4,heads=5,dim_head=32, attn_dropout=0.1,ff_dropout=0.1	0.880	0.891	0.878	0.877	0.959

SUPPLEMENTARY TABLE 4. PERFORMANCE COMPARISON OF CNN PARAMETER COMBINATIONS

Configuration	ACC	Precision	Recall	F1-score	AUC
filters=[256,128,64],kernels=[7,7,5], pool_sizes=[2,2,2],dropout=0.2, max pooling	0.879	0.893	0.871	0.876	0.960
filters=[256,128,64,32],kernels=[7], pool_sizes=[2],dropout=0.2, max pooling	0.885	0.885	0.891	0.886	0.957
filters=[256,128],kernels=[7,7], pool_sizes=[2,2],dropout=0.2, max pooling	0.885	0.897	0.876	0.883	0.960
filters=[256,256],kernels=[7,7], pool_sizes=[2,2],dropout=0.2, max pooling	0.885	0.889	0.898	0.884	0.960
filters=[128,64],kernels=[7,7], pool_sizes=[2,2],dropout=0.2, max pooling	0.884	0.891	0.883	0.883	0.960
filters=[256,128],kernels=[5,7], pool_sizes=[2,2],dropout=0.2, max pooling	0.888	0.892	0.890	0.888	0.961
filters=[256,128],kernels=[3,7], pool_sizes=[2,2],dropout=0.2, max pooling	0.887	0.894	0.896	0.886	0.960
filters=[256,128],kernels=[5,5], pool_sizes=[2,2],dropout=0.2, max pooling	0.888	0.897	0.883	0.887	0.959
filters=[256,128],kernels=[5,9], pool_sizes=[2,2],dropout=0.2, max pooling	0.886	0.892	0.884	0.885	0.960
filters=[256,128],kernels=[5,7], pool_sizes=[2,2],dropout=0.2, average pooling	0.883	0.885	0.889	0.883	0.957

SUPPLEMENTARY TABLE 5. PERFORMANCE COMPARISON OF CROSS-MODAL FUSION MODULE PARAMETERS

Configuration	ACC	Precision	Recall	F1-score	AUC
single interaction	0.885	0.893	0.880	0.883	0.960
two interactions	0.888	0.892	0.890	0.888	0.961
two interactions,Head=3,dim=16	0.887	0.897	0.881	0.885	0.960
two interactions,Head=4,dim=16	0.888	0.892	0.890	0.888	0.961
two interactions,Head=5,dim=16	0.888	0.892	0.888	0.887	0.960
two interactions,Head=4,dim=32	0.888	0.896	0.884	0.886	0.960